

Yuning Chen

☎ (209) 316-1359 • ✉ ychen372@ucmerced.edu • 🌐 ycucm.github.io • in [YC](#) • 📄 [YC](#)

RESEARCH INTEREST

I work in Machine Learning for Systems, with hands-on experience in Time Series and Causal Learning (KDD'24), Large Language Models (MLSys'24), and Generative AI (MobiSys'24).

EDUCATION

Ph.D. in Electrical Engineering and Computer Science, **University of California, Merced** 2021 - 2025 (*expected*)
Advisor: Prof. [Wan Du](#).

Bachelor in Software Engineering, **Tsinghua University** 2017 - 2021
Honor: *Outstanding Bachelor's Graduate* (10%).

PROFESSIONAL EXPERIENCE

Software Engineer Intern, Bytedance, San Jose, CA May. 2024 - Aug. 2024
Advisor: Dr. [Yongping Tang](#). Machine Learning for Systems, Network Engineering, Software-Defined Networks.

- Perform research in the field of Traffic Engineering with a focus on IP anycast, aiming to investigate innovative machine learning methods that enhance the quality of service for global TikTok DNS and CDN users.

Research Intern, Alibaba Cloud U.S., Sunnyvale, CA May. 2023 - Aug. 2023
Advisor: Dr. [Pan Hu](#), Dr. [Yunfei Ma](#). Large Language Model (LLM), Cloud Computing, IP Anycast.

- Propose a benchmark with 1011 problems for cloud-native configuration generation and a scalable evaluation platform for code assessment, and analyze the performance of 13 LLMs, offering strategies for quality and efficiency improvement.
- Build an IP anycast optimization system that formulates the AS path prepending optimization as a weighted MAX-SAT problem and propose an efficient scanning method to obtain the potential connections for global IP anycast clients.

PUBLICATIONS

(* denotes equal contribution)

- Yuning Chen, Kang Yang, Zhiyu An, Brady Holder, Luke Paloutzian, Khaled Bali, Wan Du. **MARLP: Time-series Forecasting Control for Agricultural Managed Aquifer Recharge**. ACM KDD 2024.
- Yifei Xu*, Yuning Chen*, Xumiao Zhang*, Xianshang Lin, Pan Hu, Yunfei Ma, Songwu Lu, Wan Du, Z. Morley Mao, Dennis Cai, and Ennan Zhai. **CloudEval-YAML: A Practical Benchmark for Cloud Configuration Generation**. MLSys 2024.
- Kang Yang, Yuning Chen, and Wan Du. **OrchLoc: In-Orchard Localization via a Single LoRa Gateway and Generative Diffusion Model-based Fingerprinting**. ACM Mobisys 2024.
- Kang Yang*, Yuning Chen*, Xuanren Chen, and Wan Du. **Link Quality Modeling for LoRa Networks in Orchards**. ACM/IEEE IPSN 2023.
- Yifei Xu*, Yuning Chen*, Xumiao Zhang*, Xianshang Lin, Pan Hu, Yunfei Ma, Songwu Lu, Wan Du, Z. Morley Mao, Ennan Zhai, and Dennis Cai. **CloudEval-YAML: A Realistic and Scalable Benchmark for Cloud Configuration Generation**. Machine Learning for Systems Workshop, NeurIPS 2023.
- Zhizhang Hu*, Shangjie Du*, Yuning Chen, Xuan Zhang, Wan Du, Asa Bradman, Shijia Pan. **Poster: Enhancing Fault Resilience of Air Quality Monitoring in San Joaquin Valley: A Data Equity Analysis**. ACM SenSys 2023.
- (Under Submission to NeurIPS 2024) Language to Constraints for Safe Predictive Control.
- (Under Submission to ACM IMC 2024) AnyPro: IP Anycast Optimization based on AS-Path Prepending.

SERVICES

- **Conference Reviewer:** NeurIPS'24, ACM SenSys'23 Poster&Demo, ACM MobiCom'24 AE, ACM MobiSys'24 AE.
- **Journal Reviewer:** ACM TOSN, IEEE TMC, IEEE IoTJ.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, C++, Golang, MatLab, MySQL.
- **Technologies:** PyTorch, Hugging Face, OpenCV, TensorFlow, Kubernetes, Linux, Git, Scikit-learn, Pandas, NumPy, FFMPEG, WebRTC.